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THE DEVELOPMENT AND TESTING OF VISION NAVIGATION FOR PROXIMITY OPERATIONS AND DOCKING USING THE ISS SELFIE

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Obtaining on-orbit flight-quality data for development and testing of image processing algorithms and other vision navigation techniques typically requires a significant investment of resources. The International Space Station (ISS) Selfie activity provides a collection of actual on-orbit imagery and trajectory data of docking assets viewed as the standards by the international community. The design of this activity is to provide an extensive data set to a variety of lighting conditions, nominal and off-nominal approach trajectories, operational procedures, and on-orbit camera calibrations.

The Johnson Space Center (JSC) has partnered with Draper Laboratory to extend existing terrestrial vision navigation technology for in-space applications; particularly to address the demands for autonomous rendezvous and docking (AR&D). A proposed vision navigation algorithm matches observed features to both pre-surveyed points found in a catalog and features of opportunity to determine the relative pose between two spacecraft. The ISS Selfie on-orbit testing framework introduces a cost effective approach for evaluating the relative navigation performance and robustness to the on-orbit environment.

INTRODUCTION

The development and application of vision navigation continues to expand into a variety of fields ranging from robotics, drones, cars, rovers, aircraft, situational awareness, GPS-denied environments, and so forth. Its potential use for on-orbit operations, particularly for AR&D, has been recognized for many years. As both crewed and uncrewed spacecraft continue to support deep space exploration initiatives, AR&D is a critical technology to enable emerging complex mission objectives. One elusive component essential for AR&D is having a relative navigation system that is flexible to support a variety of mission demands, but robust to the rigors of the space environment. Despite the awareness of vision navigation's potential and the great advantages it possesses in reducing navigation systems' mass, power, and volume while providing accurate relative state information, its use has not yet matched its potential. Factors that may explain why it has not been adopted more readily includes its limited maturity for space applications and its robustness to the on-orbit environment and lighting conditions. The ISS Selfie is an effort that's intended to provide a dataset for developers of vision navigation systems which addresses these factors.

Obtaining on-orbit flight heritage typically requires a large financial investment and poses significant challenges. Other options are often pursued to enhance the technology level such as generating

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computer simulated imagery or utilizing sophisticated lab testing environments (e.g. the Space Operations Simulation Center). However, even the best simulated or lab generated imagery may not accurately emulate the actual lighting conditions experienced on-orbit. As vision navigation technology matures, having on-orbit imagery to validate and quantify its performance becomes essential to reduce program risk and vulnerability.

A wide array of orbital imagery is available, but its use for development and performance assessment is limited because the camera properties and lighting conditions are typically unknown in addition to a lack of sufficiently accurate trajectory data. Existing imagery is also limited by the scope of past operations, making it difficult or impossible to test particular relative trajectory profiles under a specific set of conditions. Is there a way to develop and test the needed vision navigation algorithms for rendezvous and docking with actual on-orbit imagery without these constraints? One potential solution is to utilize the ISS and its robotic assets. The ISS Selfie activity is an on-orbit activity which uses the ISS Mobile Servicing System (MSS) to *fly* a predefined trajectory in a variety of on-orbit lighting conditions to the recently installed International Docking Adapter 2 (IDA-2), which represents the consolidation of international standards for docking systems. The trajectory data and corresponding imagery are then captured and made available for future testing.

This paper provides a brief overview of the ISS Selfie activity, highlights how it can benefit the development of AR&D technology for future space exploration programs, and demonstrates how the acquired data can be utilized to evaluate potential vision navigation algorithms performance in the on-orbit environment during AR&D operations.

THE ISS SELFIE

The main purpose of the ISS Selfie activity is to obtain imagery and trajectory data emulating the last few meters of a spacecraft's final approach to a docking system that meets the new International Docking System Standard (IDSS)[1]. The ISS Selfie gathers imagery of several unique approach trajectories and captures the relative motion intended to envelope typical docking scenarios. Each trajectory is repeated with four different lighting conditions intended to encompass the spectrum of on-orbit *day and night* cycle experienced during rendezvous operations. The ISS MSS trajectory data is recorded throughout the activity to provide an estimate of the *true* pose for subsequent performance assessment.

The ISS Selfie attempts to closely replicate the last 10 meters of the final approach of a docking spacecraft. Trajectories are developed within the constraints of the MSS and tested with the prototype MSS configuration as shown in Figure 1. Despite the confinements of the MSS with its limited reach, slow speeds, and flight operation constraints; the flexibility of the MSS provides an assortment of trajectories to be simulated quickly while capturing the imagery as if several different approach profiles were actually flown.

Currently, the only IDSS-compliant docking system on the ISS is the IDA-2 as shown in Figure 2 which is mounted on PMA-2 of Node 2 forward. As a result, the nearest mounting point for the MSS is Node 2 aft. This provides the MSS a reach of 4.5 meters from IDA-2. To increase the reach to 7.5 meters, the MSS can add the Special Purpose Dexterous Manipulator (SPDM) to the Space Station Remote Manipulator System (SSRMS) as shown in Figure 3. This is the configuration adopted for the ISS Selfie activity.

The maximum velocity of the MSS is less than 6 cm/s. While this may appear to be operationally burdensome with longer run times, it captures the final approach and docking rates that are typically

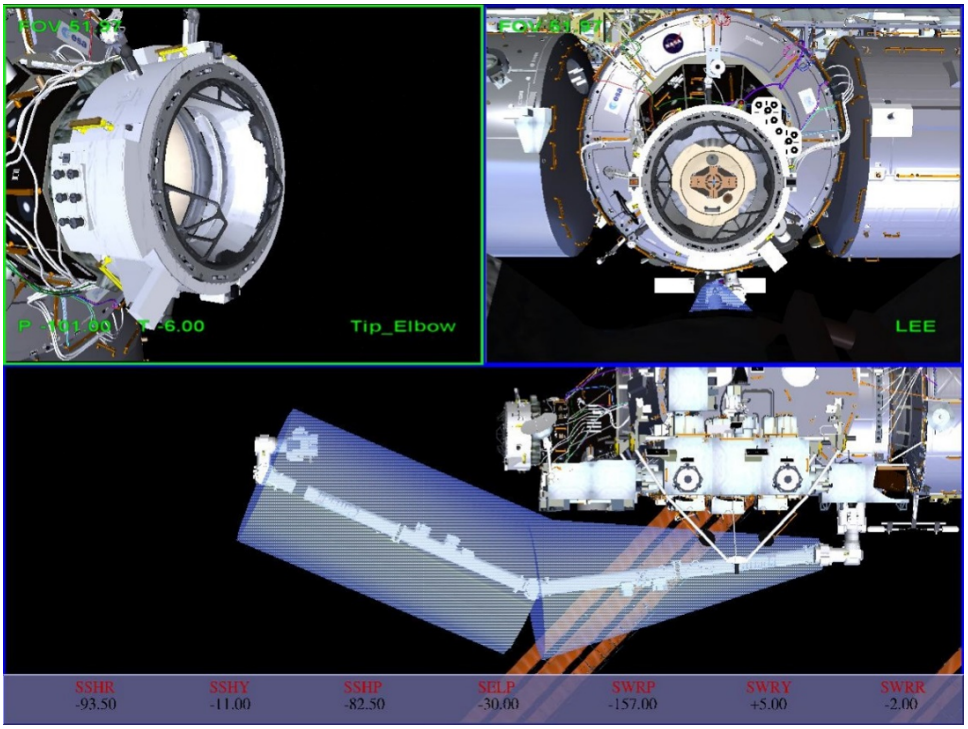


Figure 1. Sample of Computer Generated Imagery for ISS Selfie Demonstration

between 3-6 cm/s. Besides the limitation of the MSS, flight rules regarding MSS velocity and structural clearance, joint angle restrictions, and other operational stipulations impact potential relative motion profiles. The ISS Selfie trajectory design accounted for these constraints of the ISS MSS.

ISS Selfie Objectives The initial objectives for the camera's relative motion to the docking port included the following; an approach from at least 10 meters away, a roll of 180 degrees, a profile along a feasible approach corridor, an off-axis trajectory to emulate a non-centerline docking camera, approaches along both the Centerline Docking Target (CDT) and the Peripheral Docking Target (PDT), a tracking pass (ensuring the camera is pointed at a single feature throughout the motion) with combined rotational and translational motion, lateral motion, varying rates, perturbations to represent thruster firings, velocity profile between 3-6 cm/s to simulate docking rates, approach velocities greater than 3-6 cm/s, in-plane and cross-track motion, lateral and rotational offsets in all axes, and coming to a complete stop before resuming motion to emulate a hold point during approach. Additional goals for the ISS Selfie trajectory design and MSS configuration included keeping the entire IDA-2 along with the PDT and hemispherical reflectors at the maximum achievable range within the camera's field of view, having the CDT backplate fill the camera's field of view at the minimum range, and ensuring the resulting motion is repeatable and limited in duration to be captured within a specific lighting condition.

Another test objective is to capture the following lighting conditions: *midnight* or the eclipse period of the orbit, *sunrise* as the vehicle transitions from darkness to light, *noon* or peak lighting, and *sunset* to capture the transition from light to darkness. The approximate 45 minute *day/night* cycle due to the ISS orbital period influences the selected duration of the trajectory design. In order

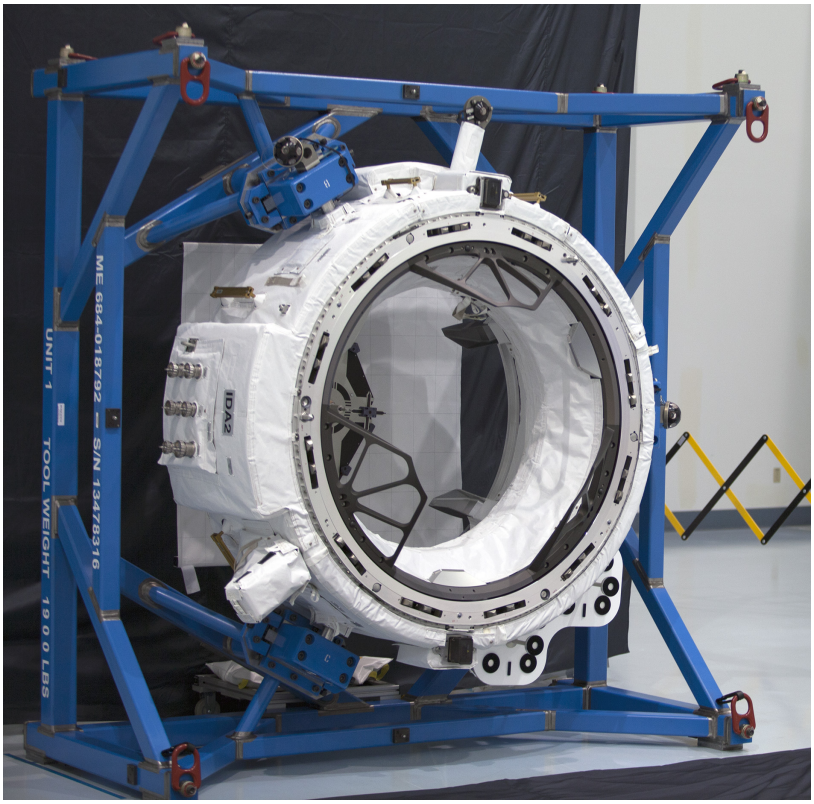


Figure 2. IDA-2 in the Space Station Processing Facility (Kennedy Space Center). IDA-2 is pictured 90 degrees clockwise from its installed orientation. Note the partially obscured CDT near the center of the image and PDT near the bottom right.

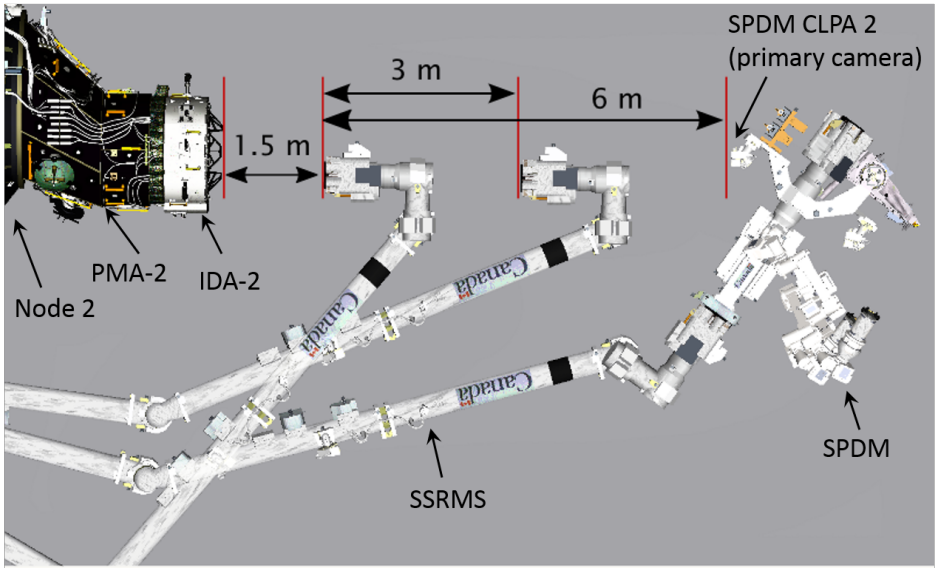


Figure 3. Minimum Distance from the SSRMS to IDA-2 and Maximum Reach from the SSRMS Alone and the SSRMS with SPDM

to have consistent lighting characteristics for each proposed segment, the duration of the relative motion profile is designed to be less than 15 minutes.

To ensure repeatability, a single trajectory is developed to incorporate all the previously defined goals while conforming to the constraints of the MSS. This helps reduce operator burden and run time while maximizing the amount and quality of the data within the allotted MSS time resources. The JSC engineering ISS robotics group utilized the Systems Engineering Simulator (SES) at JSC to simulate the relative motion dynamics and corresponding graphics rendered with Engineering DOUG Graphics for Exploration (EDGE) to evaluate and ultimately determine the trajectory profile that satisfied the ISS Selfie objectives while complying with the imposed constraints of the MSS. As shown in Figure 4, the SES is a real-time, in-the-loop engineering simulator for ISS and advanced programs. The delivered flight-like data and imagery products not only supports the trajectory development but they also help establish the necessary interfaces and formats imposed on the vision navigation algorithms that may process the data.

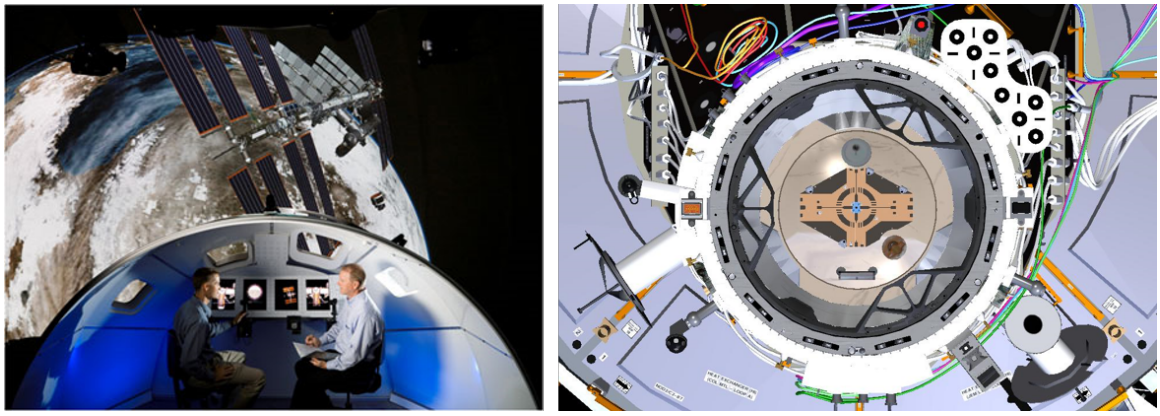


Figure 4. SES (left) and EDGE Graphics (left background, right)

The final trajectory consists of a set of 13 motion segments as depicted in Figure 5. The details regarding each step in the profile are outlined as follows:

1. Move in from farthest approach to closest approach along the centerline of the CDT
2. Move laterally starboard and nadir and re-orient the camera to point at the center of the CDT
3. Retreat along corridor
4. Track CDT center, camera moves to the center of the CDT in the y axis and zenith of the CDT
5. Re-orient the camera boresight to be parallel to the CDT centerline
6. Move in from farthest approach to closest approach along corridor, end above the CDT center
7. Translate directly nadir across CDT face
8. Translate directly zenith to center of CDT
9. Retreat along centerline in z axis and slightly to port in y
10. Retreat along centerline in z axis and slightly to starboard
11. Retreat along centerline in z axis and slightly to port in y, end at centerline in y and z axes
12. Roll approximately 30° degrees and back
13. Reset to start position

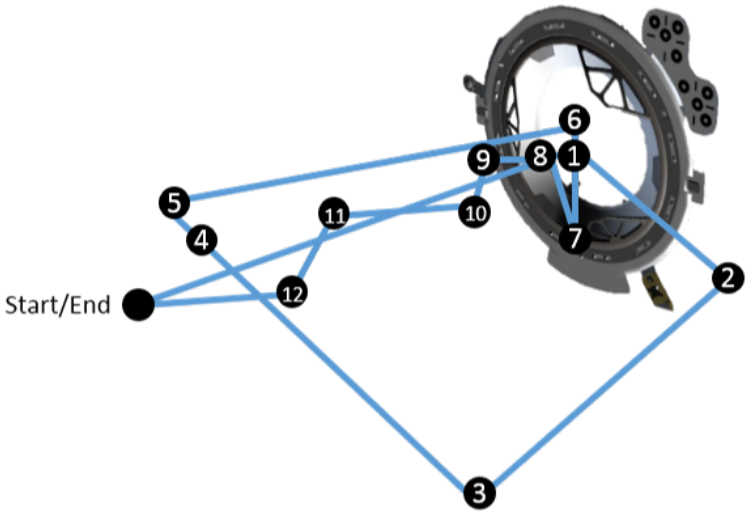


Figure 5. ISS Selfie Motion Segments

Figure 6 summarizes the goals, denoted as requirements in the figure, that are accomplished with the final trajectory design. Red indicates the segment does not meet the goal, yellow suggests the goal is partially met, green shows the segment satisfies the goal, and N/A implies the goal is not compatible with the trajectory segment.

Requirement	Segments										
--	Start-1	2-3	3-4	5-6	6-7	7-8	8-9	9-10	10-11	11-12	12-13
Centerline approach											
"Fly the corridor"											
"Tracking pass"											
Orthogonal camera orientation to target reference with position laterally offset for all range											
Orthogonal camera orientation to target reference with changing lateral offset as a function of range											
'Flying to the target from an offset' (non-orthogonal, lateral offset as a function of range)											
Representative docking approach rates											
Change rates (start-stop)											
Some lateral translation at line-of-sight normal rate (features move quickly in the frame)											
Large roll (up to 180 deg)											
Misalignment in all angles and laterally at the same time											
In-plane / cross track motion											
Rates greater than nominal docking rates											
Thruster-like perturbation											
10m range											
IDA in view at max range					N/A	N/A	N/A	N/A	N/A	N/A	
CDT backplate fills FOV at close range	N/A		N/A					N/A	N/A	N/A	N/A

Figure 6. ISS Selfie As-Flown Trajectory vs. Requirements

ISS Selife Activity Once the trajectory had been finalized, SES simulation showed that it would take a flight control team approximately one hour to complete a single run due to the standard control practices involving the MSS. The MSS is typically commanded from the ground point-by-

point, with each node being manually entered, commanded, and the system coming to a complete stop after each specified location. Another control mode for the MSS exists, although it is rarely used. This mode allows for a set of values to be sent in a single command, driving the system through each point in the set without stopping. After adding this rarely-used control mode to the MSS simulation, the trajectory could be completed in approximately 15 minutes, 45 minutes faster than the other control mode.

The ISS Selfie was formally approved by various ISS control boards between 2015-2016 and was taken from concept to commands and procedures by the JSC engineering and flight operations robotics groups and the MSS manufacturer, the Canadian Space Agency. The IDA-2, the first IDSS compliant docking adapter on station, was launched in July 2016 and installed August 2016. The ISS Selfie activity took place on 9-10 November 2016 and collected more than 200 gigabytes of data and imagery comprising of more than 49 hours of video.

The activity completed four *sunrise* runs (two CDT and two PDT), five *noon* passes (two PDT and three CDT), five *sunset* runs (three PDT and two CDT), four *midnight* runs (two PDT and two CDT), and a set of imagery that could be used for camera calibration. At a minimum, each of the 18 trajectories has four videos and one set of telemetry. One video is from the SPDM body camera centered on the target, another is from the nearby SPDM body camera, and the last two videos are the same as the previous two but with overlaid timestamps. For some passes, camera footage is available from the Japanese Experiment Module's (JEM) external cameras, which provides a *3rd person* view of the activity.

In total, 97 videos are available, each having NTSC format and encoded as mp4 video files. Due to the final configuration of the SPDM, the videos from the SPDM body cameras are rolled approximately 180 degrees, making the imagery look *upside down*. Some videos are continuous and seamless, but many are merged together from onboard and ground recordings due to loss of the ISS high bandwidth communication signal. For the entire activity, the zoom level is fixed so that the focal length is 32.4 mm, which results in a 15.218 degree field of view. Lights on the camera assemblies are left on throughout the activity. Each light is rated to provide a 52 degree cone illumination and 0.72 foot-candles of intensity at 33 feet. The sun's beta angle is approximately a negative 15 degrees. Associated telemetry is available at three different points on the SPDM: 1) the SPDM Latching End Effector (LEE), 2) body camera 1 eye point, and 3) body camera 2 eye point. The telemetry files are comma delimited csv-files that describe the position (cm) and attitude (deg) of the point they reference in the ISS Analysis Coordinate System (ISSACS) at an interval of 1Hz with an associated GMT time stamp.

In general, feature visibility is good within the videos. Most features larger than a few millimeters are apparent even when the MSS is at its furthest extension. The ISS Selfie activity captures the variable lighting conditions anticipated for on-orbit operations. In certain instances, shadows can be so dark that few, if any, features are visible within them. In others, bloom can remove features from view. Typically, certain markers and cues can be detected throughout each profile. Although objects in shadows are usually visible, shadows cast a stark contrast and move across the target surfaces quickly. The transition to sunset has a rather dramatic change approximately 15 seconds before total darkness. Once the ISS is completely eclipsed, the camera lights become apparent and the scene brightens due to the artificial light. However, the image is grainy when MSS is at its furthest extension. The video improves noticeably when the relative distance between the camera and target is reduced.

VISION NAVIGATION ANALYSIS FOR AR&D USING ISS SELFIE DATA

To demonstrate how the ISS Selfie data provides a valuable resource to evaluate the performance and robustness of vision navigation technology to the on-orbit environment for AR&D, a segment from a single run of the ISS Selfie profile shown in Figure 7 is adopted that replicates a final approach trajectory of a visiting vehicle. The incoming spacecraft is assumed to be on the V-bar such that the relative distance between the docking camera mounted on the approaching spacecraft is initially 5.5 meters in front of the baseplate of IDA-2 mounted on the forward Node 2 port. The vehicle approaches ISS at a rate of 3.3 cm/sec until the camera is within 3 meters. During this shortened segment, the vehicle holds a nearly fixed relative attitude to the docking port.

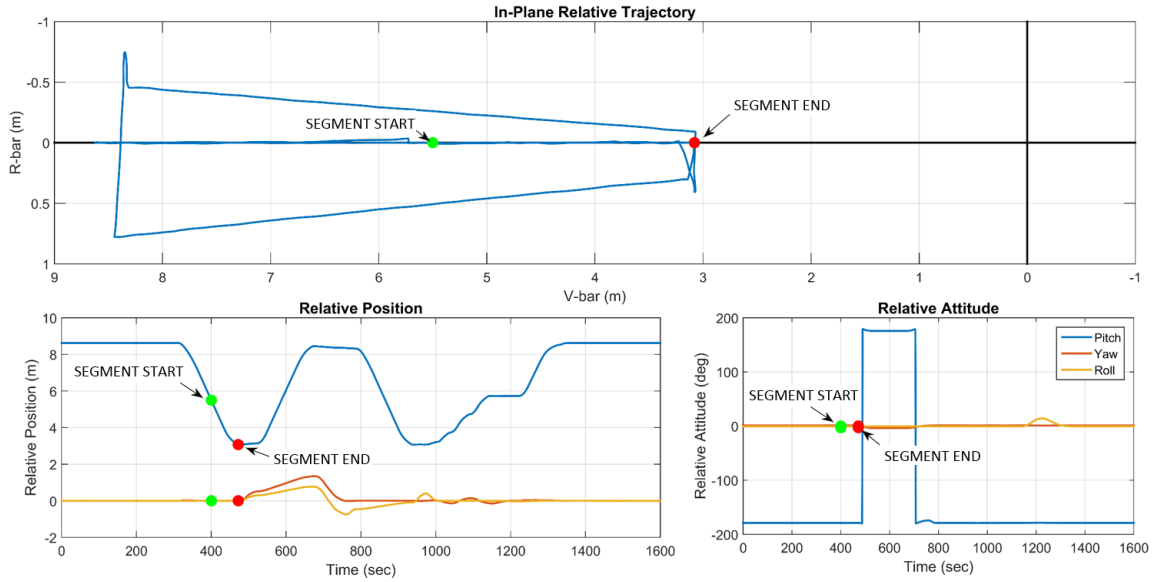


Figure 7. MSS Trajectory Data from a Single ISS Selfie Run

To evaluate the relative navigation performance of a vision system, both a simulated imagery data set and the actual on-orbit images are utilized as highlighted in Figure 8 and Figure 9 respectively. In each figure, the camera's view of the target is captured at 0.5 meter increments starting from 5.5 meters to the final 3 meter destination. Although significant detail is captured in the simulated imagery, the actual on-orbit images capture the distinct effects of on-orbit lighting, camera defects, and offsets. The contrast between the two data sets helps provide insight into the discrepancies likely to be observed between a simulated and on-orbit environments.

The simulated ISS Selfie is performed with computer generated imagery from the perspective of the SPDM camera approaching the IDA mounted on Node 2 of the ISS as shown in Figure 10. The camera's horizontal field-of-view (HFOV) is 15 deg with a pixel resolution of 720 x 540. The actual on-orbit imagery is captured with a camera whose focal length is 32.40 mm producing a 15.218 deg HFOV and image resolution of 720 x 540 pixels.

Vision Navigation Overview The implemented vision navigation algorithm was originally created by Madison et. al. [2, 3] and has extensive heritage over a variety of terrestrial programs. JSC has partnered with Draper Laboratory to extend this existing terrestrial vision navigation technol-

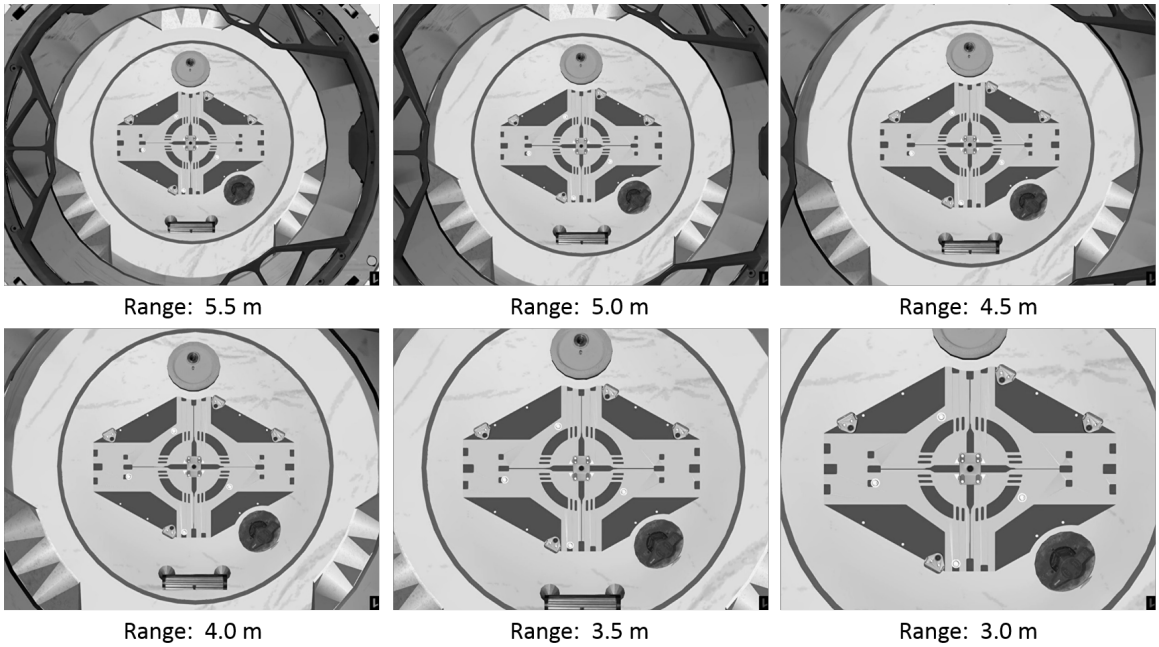


Figure 8. ISS Selfie SPDM Simulated Images as Processed by Vision Navigation Algorithm

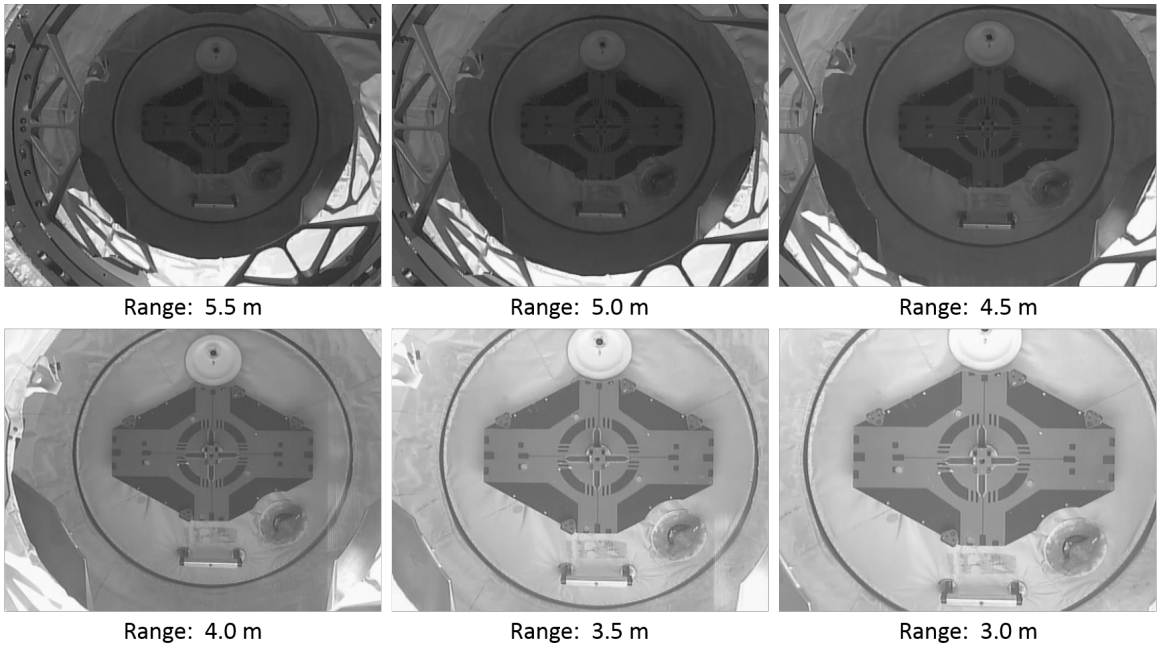


Figure 9. ISS Selfie SPDM On-Orbit Images as Processed by Vision Navigation Algorithm

ogy for in-space applications, particularly for rendezvous and docking. It detects and tracks unique Scale Invariant Feature Transform (SIFT) features [4] by first observing the intensity of each pixel and searching for gradients between pixels in a local area. The features are designed to be invariant to image scaling and rotation, and partially invariant to change in illumination and camera view.

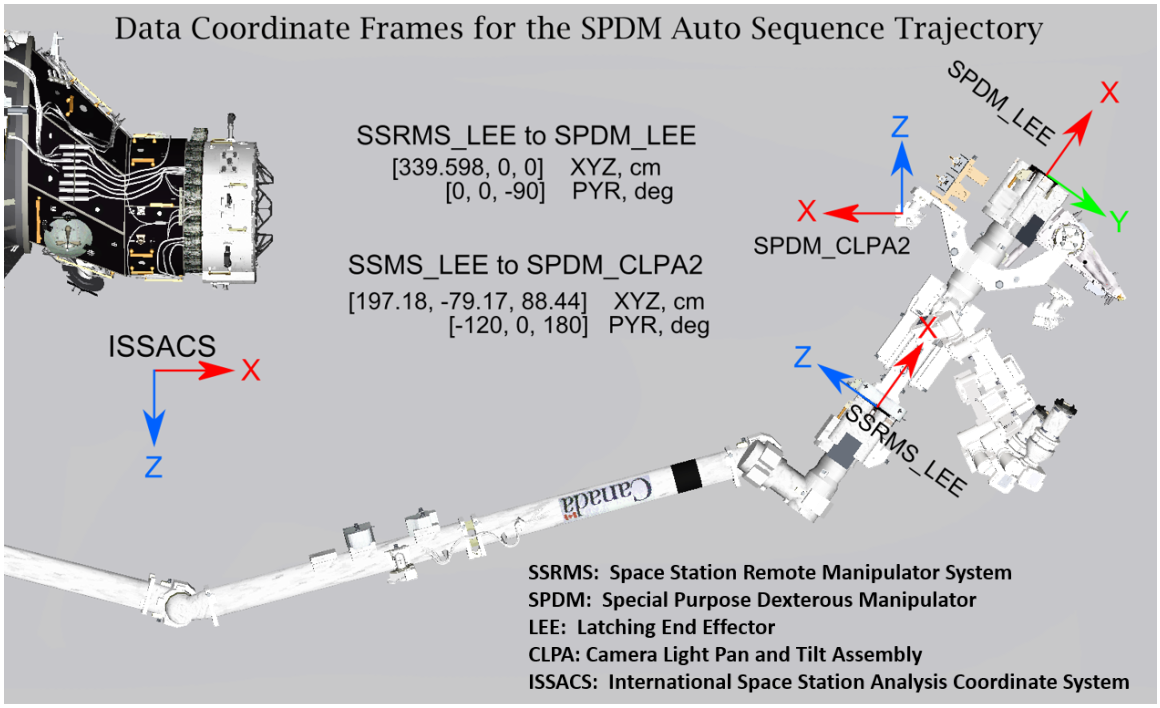


Figure 10. ISS Selfie Reference Frames

As highlighted in Figure 11, large number of features can be extracted from an image but typically a threshold is set to limit the total number to a predetermined number to reduce unnecessary computational load. Each feature has a unique gradient pattern and contrast[5] that is captured as the feature's descriptor, a 128 element vector of image gradients (16 grids x 8 gradients per grid).

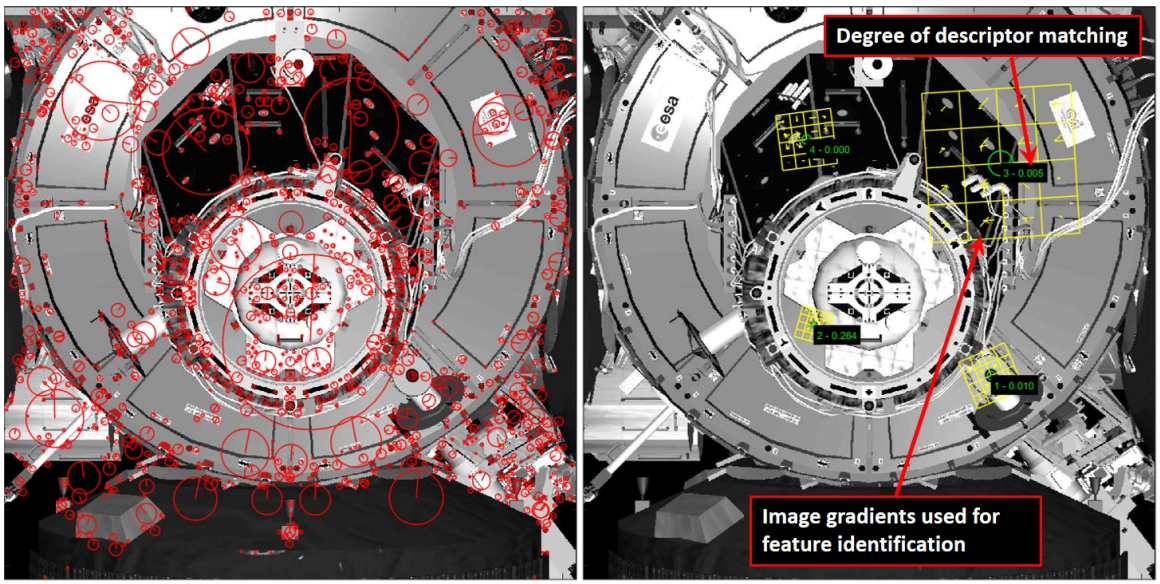


Figure 11. SIFT Features and Descriptor

This descriptor allows the algorithm to identify and match the corresponding feature in subsequent images. In addition to SIFT features, the algorithm can also process the center of various templates which provides robustness to varying lighting conditions. Sample templates that can be tracked include the CDT baseplate, docking cross, and various reflectors associated with the CDT as shown in Figure 12. Utilizing sub-templates within a predetermined main template enhances the algorithm's performance by limiting the search region for subsequent templates and ensuring an adequate number of *known features* are tracked to reliably determine the relative pose.

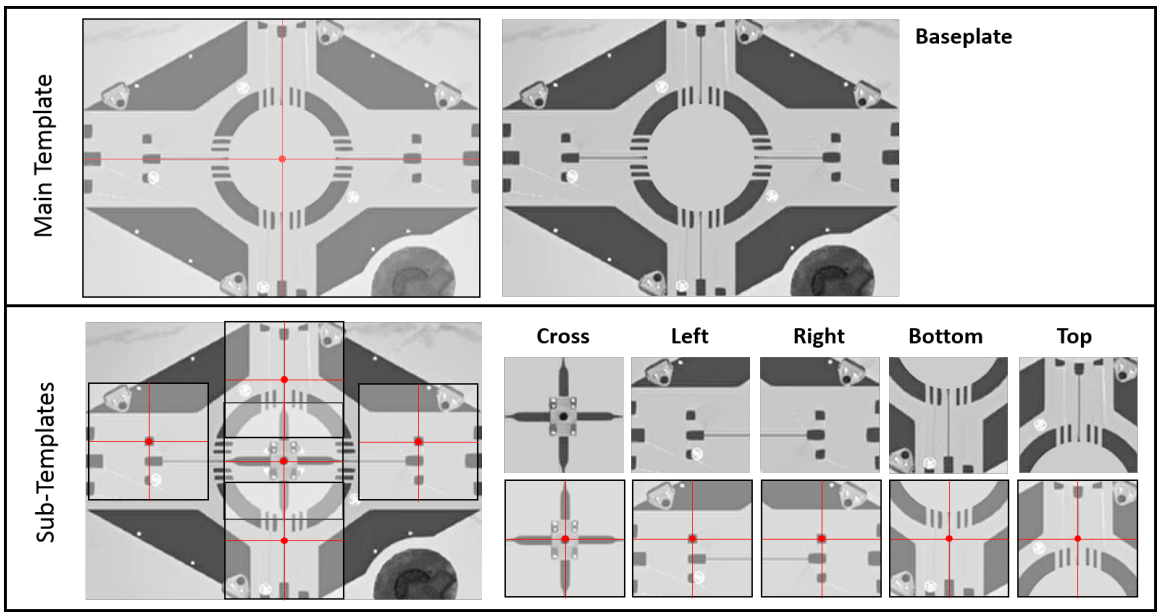


Figure 12. Main Template and Sub-templates

Relative Navigation Performance Results The resulting relative navigation performance for both data sets is captured in Figure 13 and Figure 14. The top right plot shows the actual in-plane relative trajectory (solid blue), the simulated (dashed green) segment, and the estimated trajectory based of the vision navigation algorithm (solid red). The other sub-plots on the right half of the figure show the relative position and relative attitude errors as a function of the camera's down-range distance to the CDT baseplate. The image plot on the left of the figure highlights the tracked templates and unknown features.

For the simulated data set, the final relative position error is less than 3 cm and the relative attitude error is between 0.3 deg. The relative navigation performance for the actual on-orbit imagery had an accuracy of approximately 5 cm at the final point in relative position and 0.5 deg in relative attitude. These preliminary results from a subset of a single approach profile are not intended to quantify the anticipated performance for a completely developed vision navigation system, but demonstrate the feasibility and potential utilization of the ISS Selfie data in analyzing a vision navigation system for AR&D applications.

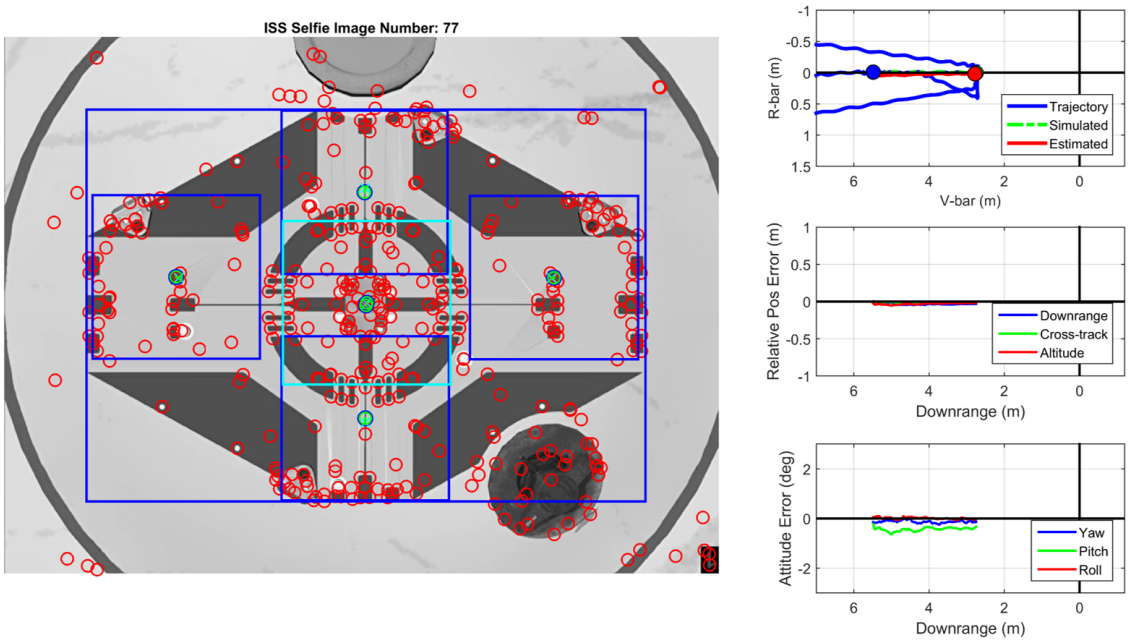


Figure 13. Relative Navigation Performance Results using Simulated Imagery

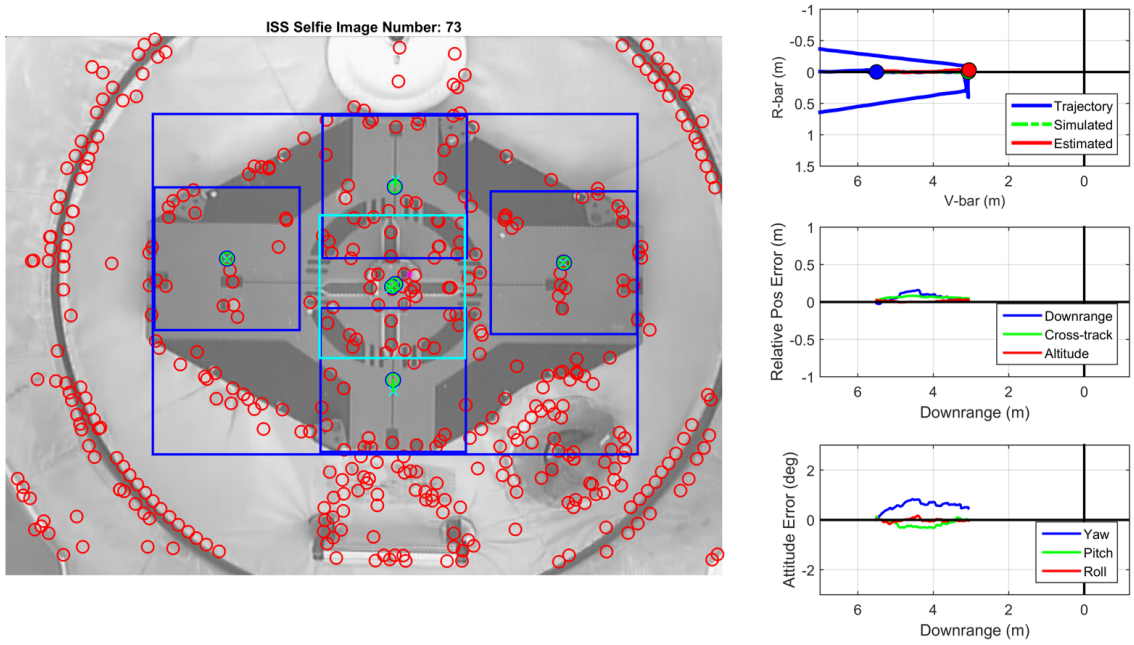


Figure 14. Relative Navigation Performance Results using On-Orbit Imagery

CONCLUSION

The ISS Selfie activity has opened another avenue for maturing the technology to support AR&D that will play a vital role for current and future deep space exploration programs. In particular, it provides an extensive data set of both on-orbit imagery and the corresponding relative state informa-

tion to evaluate the performance of potential vision navigation algorithms designed to support the final approach and docking flight phase. It captures the extreme lighting conditions experienced on-orbit and the anticipated flight profiles of vehicles rendezvousing with assets equipped with docking systems conforming to the international docking standards.

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